

# VIRTUALIZATION TECHNIQUES & TYPES



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# AGENDA

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Introduction to Virtualization

Types of Virtualization

How does Virtualization help Cloud providers

Issues & Challenges

NFV - VNF



# VIRTUALIZATION

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Foundational technology in cloud computing

Involves creating a virtual version of something

Operating System

Server

Storage device

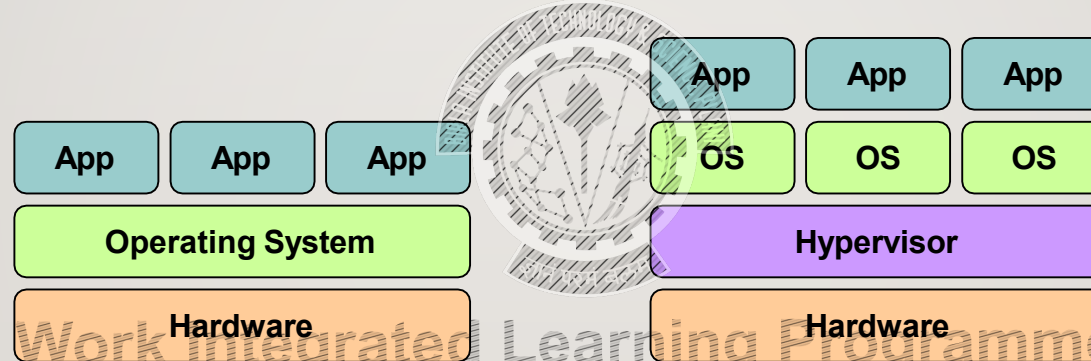
Network resources



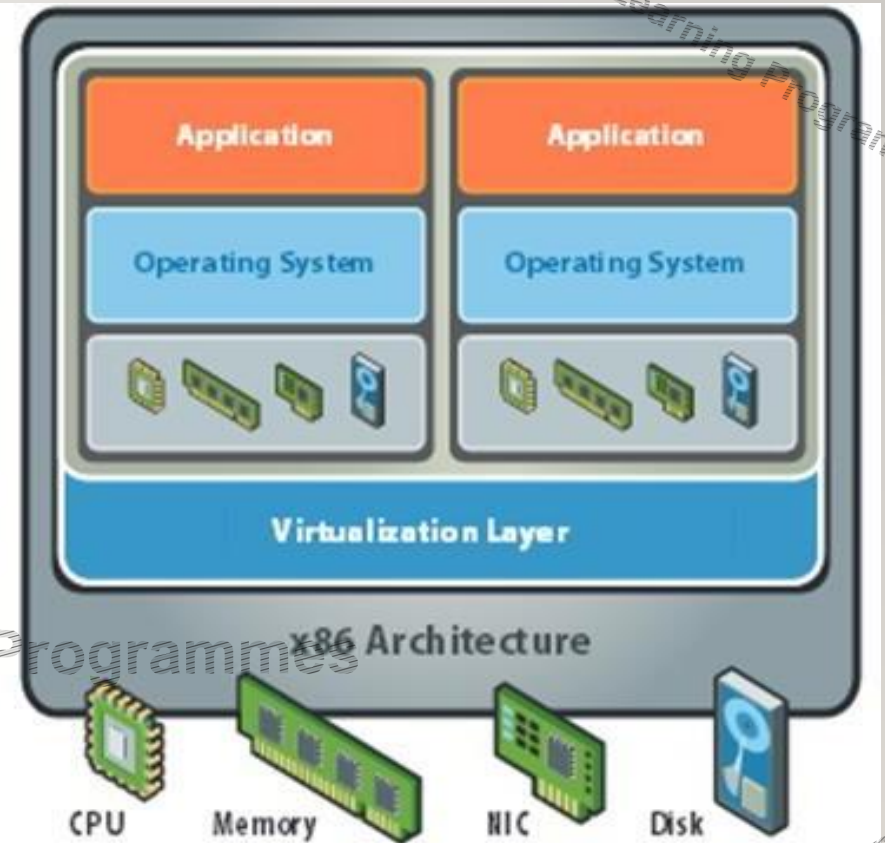
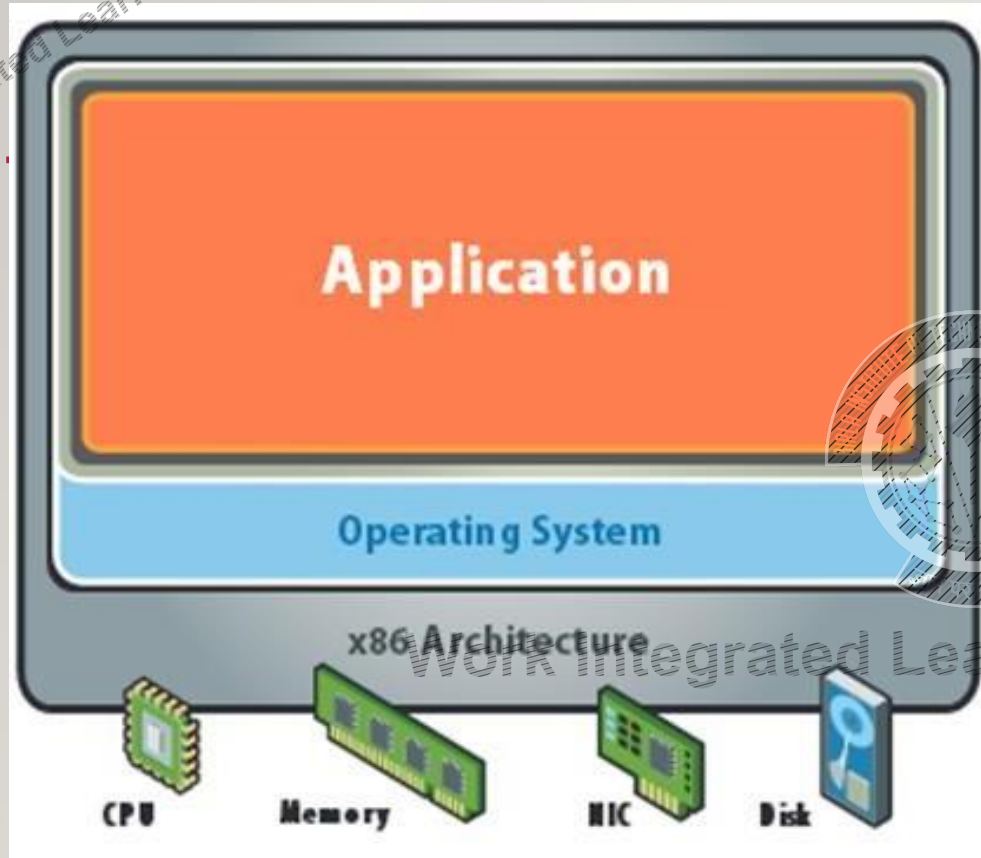
- Foundation
- Abstraction

Achieved by using software to simulate hardware functionality and create a virtual computer system.

# Key Technology is Virtualization



Virtualization plays an important role as an enabling technology for datacentre implementation by abstracting compute, network, and storage service platforms from the underlying physical hardware



# WHAT DOES VIRTUALIZATION DO?

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- Virtualization allows multiple operating system instances to run concurrently on a single computer
- It is a means of separating hardware from a single operating system.
- Each “guest” OS is managed by a Virtual Machine Monitor (VMM) also known as a hypervisor.
- Because the virtualization system sits between the guest and the hardware, it can control the guests’ use of CPU, memory, and storage, even allowing a guest OS to migrate from one machine to another.
- Instead of purchasing and maintaining an entire computer for one application, each application can be given its own operating system, and all those operating systems can reside on a single piece of hardware.
- Virtualization allows an operator to control a guest operating system’s use of CPU, memory, storage, and other resources, so each guest receives only the resources that it needs.

# CHANGES AFTER VIRTUALIZATION

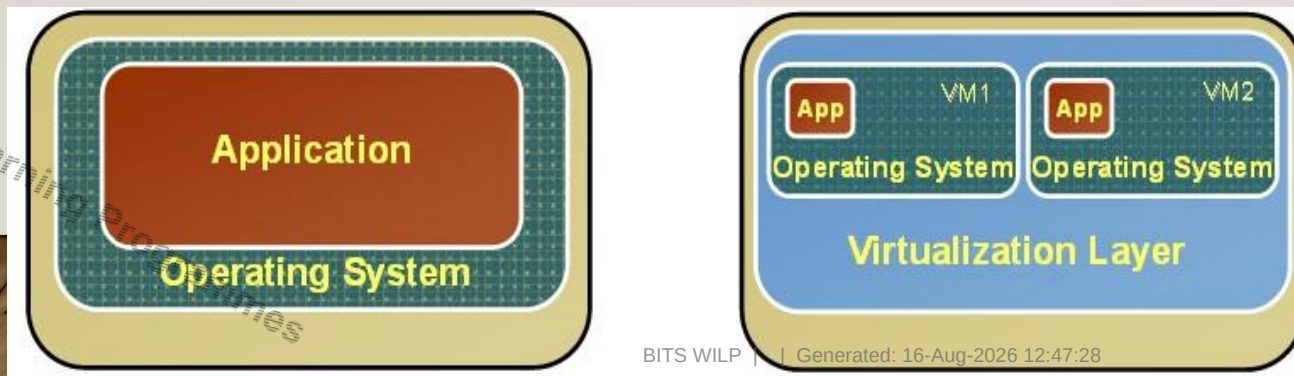
## Before Virtualization

- Single OS image per machine
- Software and hardware tightly coupled
- Running multiple applications on same machine often creates conflict
- Underutilized resources
- Inflexible and costly infrastructure



## After Virtualization

- Hardware-independence of operating system and applications
- Virtual machines can be provisioned to any system
- Can manage OS and application as a single unit by encapsulating them into virtual machines



# VIRTUALIZATION ARCHITECTURE

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- Guest OS assumes complete control of the underlying hardware.
- Virtualization architecture provides this illusion through a hypervisor/VMM.
- Hypervisor/VMM is a software layer which:
  - Allows multiple Guest OS (Virtual Machines) to run simultaneously on a single physical host
  - Provides hardware abstraction to the running Guest OS' and efficiently multiplexes underlying hardware resources



# HYPERVISOR – BARE METAL & HOSTED

**Bare-metal hypervisor:** Runs directly on the physical hardware without an underlying operating system.  
**Hosted hypervisor:** Runs on top of an operating system, which in turn runs on the physical hardware.



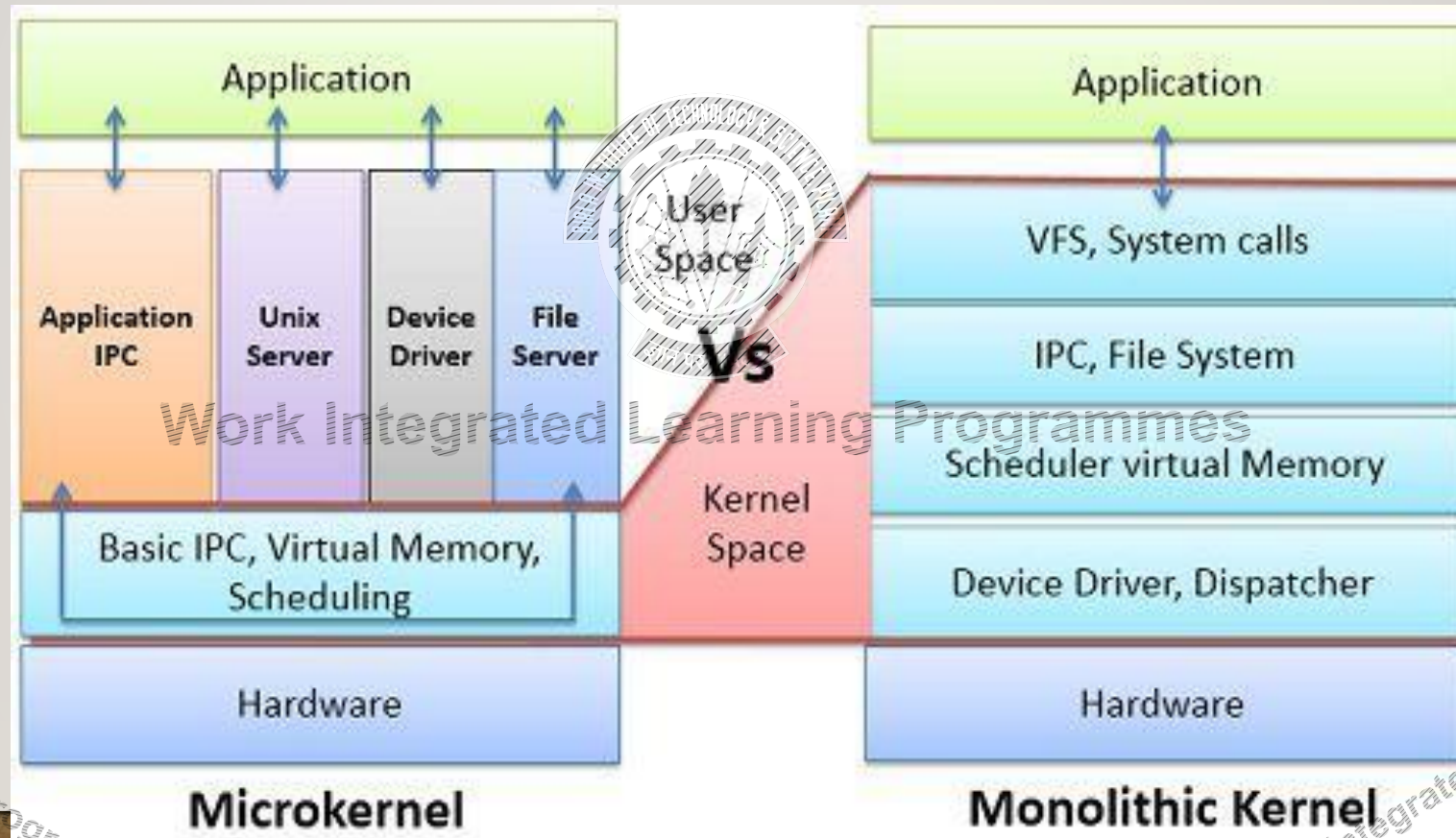
# HYPERVISOR DESIGN GOALS

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- Isolation
  - Security isolation
  - Fault isolation
  - Resource isolation
- Reliability
  - Minimal code base
  - Strictly layered design
  - Not extensible
- Scalability
  - Scale to large number of cores
  - Large memory systems

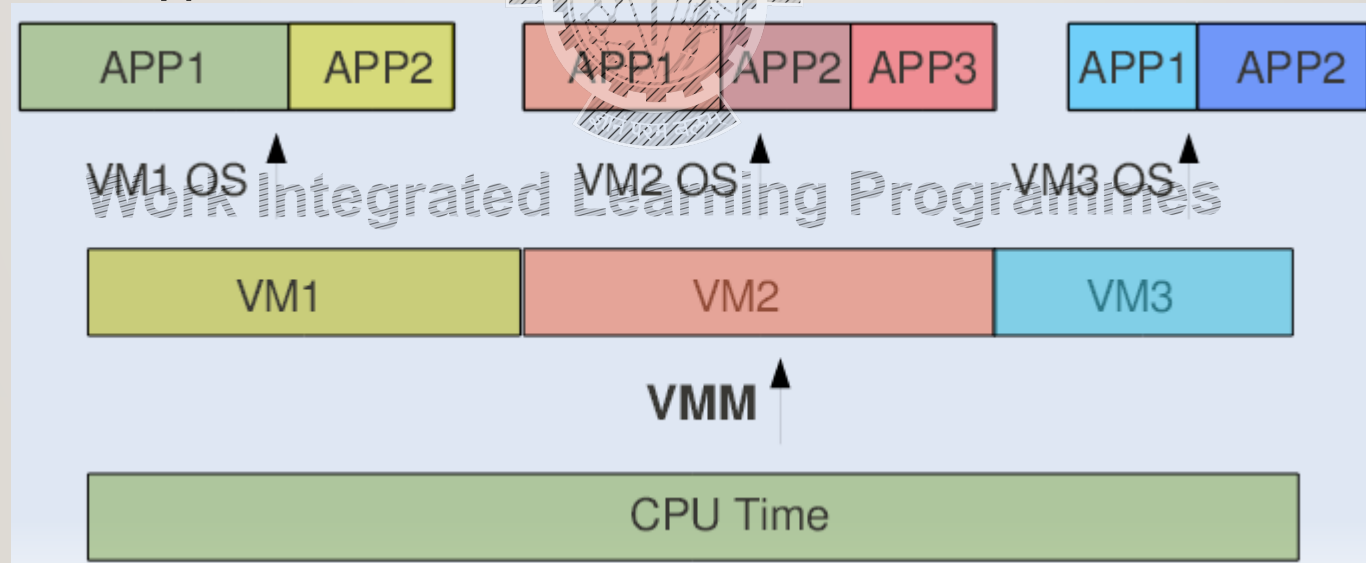


# HYPERVISOR MONOLITHIC VERSUS MICROKERNELIZED



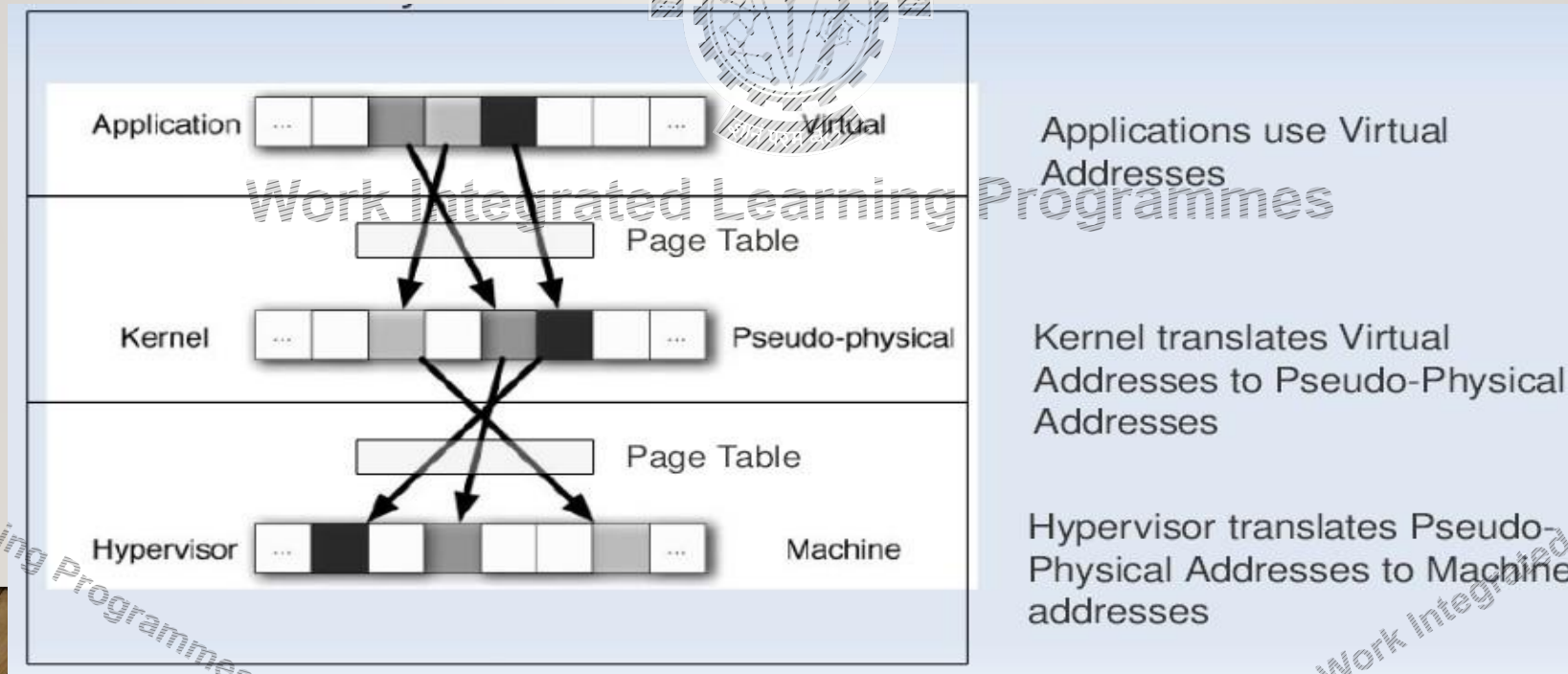
# CPU SHARING

- VMM or Hypervisor provides a virtual view of CPU to VMs.
- In multi processing, CPU is allotted to the different processes in form of time slices by the OS.
- Similarly VMM or Hypervisor allots CPU to different VMs.



# MEMORY SHARING

- In Multiprogramming there is a single level of indirection maintained by Kernel.
- In case of Virtual Machines there is one more level of indirection maintained by VMM



# IO SHARING

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- Direct I/O (DIO)

**Direct access:** Allows the guest operating system to access I/O devices directly, bypassing the hypervisor.

**Performance:** This can improve performance, especially for I/O-intensive workloads.

**Security:** Requires careful management to prevent security risks



- Pass-through:

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**Dedicated device:** Assigns a physical I/O device exclusively to a single VM.

**High performance:** Ideal for VMs with specific I/O requirements, like high-performance computing or real-time applications.

**Limited flexibility:** Reduces the flexibility of resource allocation.

# IO SHARING

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- Shared I/O

**Multiple VMs:** Multiple VMs sharing a single I/O device.

**Resource management:** Hypervisor manages access to the shared device. Ensures fair allocation & conflicts.

**Performance trade-offs:** Performance overhead when there are resource contentions.



- Virtual I/O Device

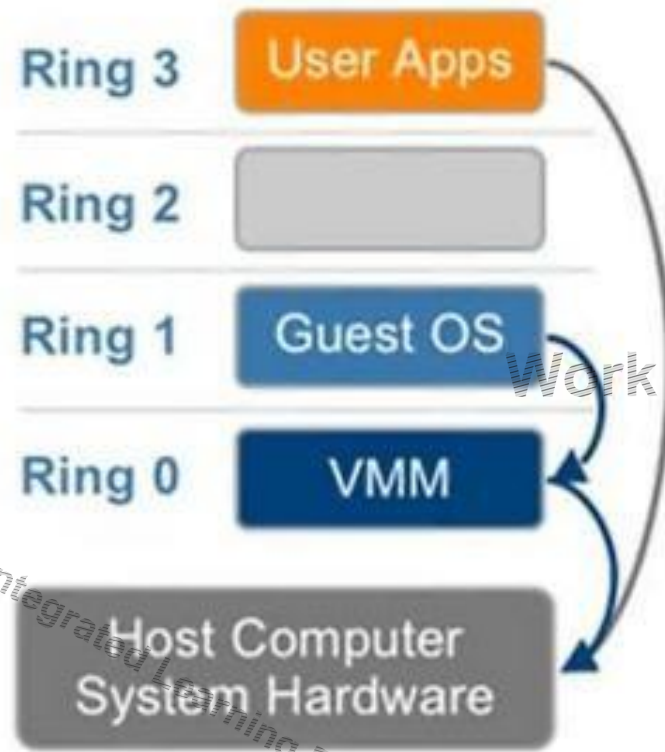
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- **Emulated devices:** Creates virtual I/O devices that are presented to the VMs as physical devices.
- **Flexibility:** Offers greater flexibility in resource allocation and management.
- **Performance:** May have some performance overhead compared to direct I/O

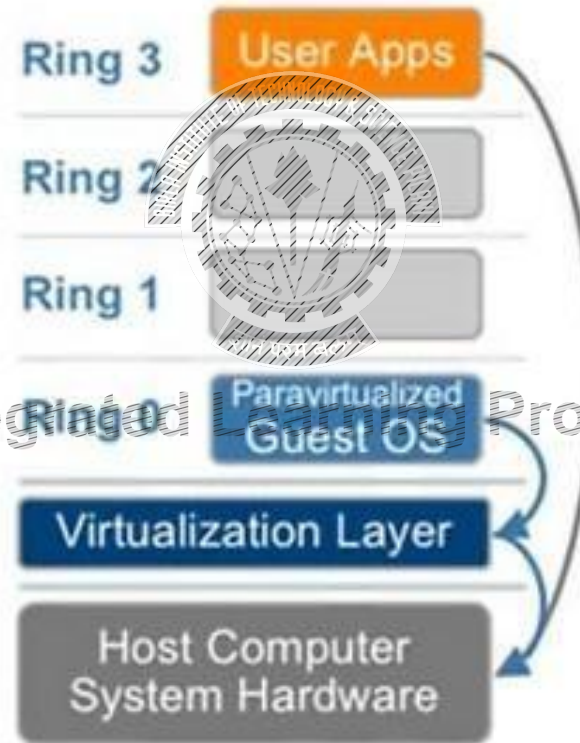
- **Paravirtualization** is virtualization in which the guest operating system (the one being virtualized) is aware that it is a guest and accordingly has drivers that, instead of issuing hardware commands, simply issue commands directly to the host operating system. This also includes memory and thread management as well, which usually require unavailable privileged instructions in the processor.
- **Full Virtualization** is virtualization in which the guest operating system is unaware that it is in a virtualized environment, and therefore hardware is virtualized by the host operating system so that the guest can issue commands to what it thinks is actual hardware, but really are just simulated hardware devices created by the host.
- **Hardware-Assisted Virtualization** is a type of Full Virtualization where the microprocessor architecture has special instructions to aid the virtualization of hardware. These instructions might allow a virtual context to be setup so that the guest can execute privileged instructions directly on the processor without affecting the host. Such a feature set is often called a Hypervisor.

# Architectural Comparison

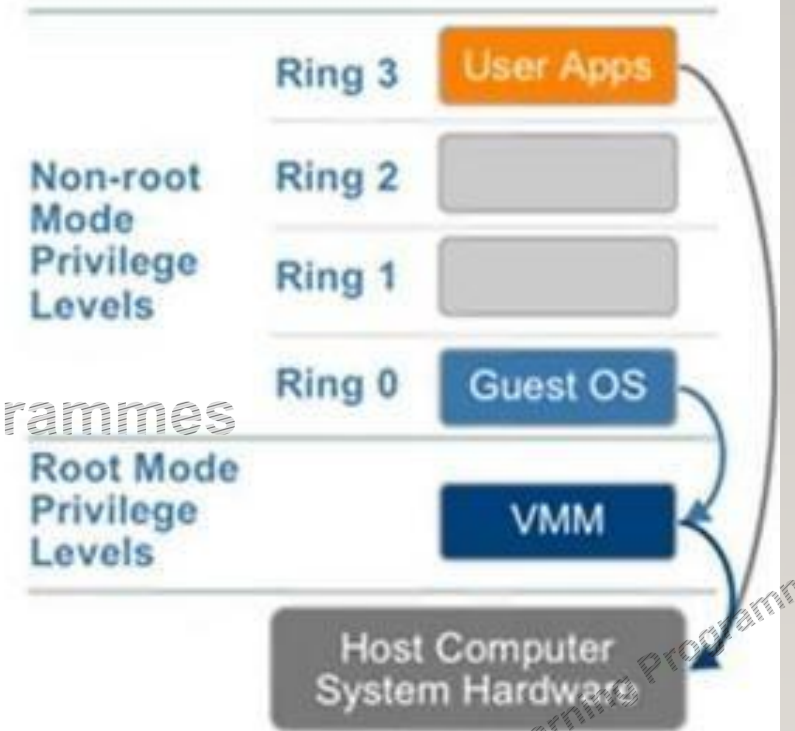
## Full Virtualization



## Paravirtualization



## Hardware Assisted



# HARDWARE & SOFTWARE VIRTUALIZATION

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Techniques used to create multiple virtual environments on a single physical system.

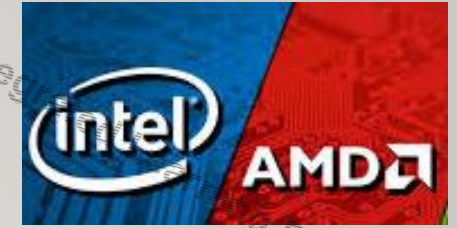


Differ in their underlying mechanisms and applications.

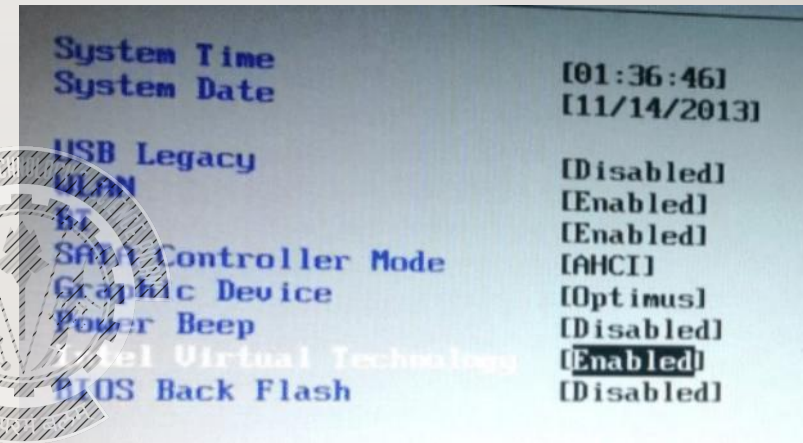
- Mechanism
- Performance
- Flexibility
- Cost

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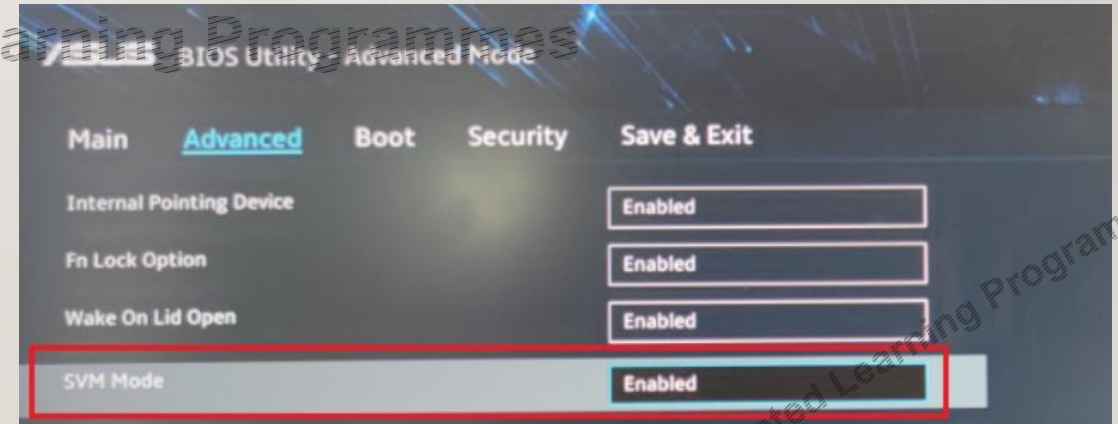
# HARDWARE VIRTUALIZATION



Mechanism: Computer Hardware directly supports the creation of multiple virtual machines. The processor itself provides the capability to run multiple operating systems simultaneously.



Examples: Intel VT-x and AMD-V are examples of hardware virtualization technologies. These technologies allow a physical CPU to emulate multiple virtual CPUs.



Many ways to enable the virtualization – Secure Virtual Machine setting or Virtual Tech setting

# SOFTWARE VIRTUALIZATION

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**Mechanism:** Software creates the illusion of multiple virtual machines. A hypervisor, a special type of software, manages the allocation of resources like CPU, memory, and storage to each virtual machine.

**Example:** VMware, VirtualBox, and Hyper-V are popular software virtualization platforms. They use software techniques to divide a physical computer's resources among multiple virtual machines.



# HARDWARE & SOFTWARE VIRTUALIZATION

|                    | Hardware Virtualization                       | Software Virtualization                                          |
|--------------------|-----------------------------------------------|------------------------------------------------------------------|
| <b>Mechanism</b>   | Hardware-based                                | Software-based                                                   |
| <b>Performance</b> | Generally faster due to hardware support      | Can be slower due to software overhead                           |
| <b>Flexibility</b> | Limited by hardware capabilities              | More flexible due to software control                            |
| <b>Cost</b>        | Requires hardware with virtualization support | Can be less expensive as it doesn't require specialized hardware |

## UseCase

Hardware Virtualization: Server consolidation, running multiple operating systems on a single physical server.

Software Virtualization: Desktop virtualization, providing virtual desktops to users, application testing, and cloud computing.

# HOW DOES VIRTUALIZATION HELP CLOUD PROVIDERS?

**Efficient Resource Utilization:** When a VM is stopped in a Cloud like AWS, its resources, including CPU cores, memory and storage are deallocated.

These resources are then available for other VMs to use.

**Preserving the State:** While the resources are released, the state of the VM is preserved. This includes the operating system, applications, and data. This allows the VM to be started again quickly without having to reinstall everything.

The stopped VM retains its configuration and state, allowing it to be started again without losing any data.

This approach ensures that stopped VMs do not consume unnecessary resources while still preserving their state for future use.

Preserving the state of a VM is an internal mechanism to store configuration and Data.  
This cannot be equated to/considered as image creation

# HOW DOES VIRTUALIZATION HELP CLOUD PROVIDERS?

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**Server Consolidation:** Multiple VMs in one single physical server – reduce hardware costs, energy efficiency

**Scalable and Flexible**

Rapid provisioning : Quick creation and deletion of VMs based on demand

Customization: Wide range of VM configurations

**Cost Effectiveness : Pay-as-you-go, Reduced CapEx**

Mutli Tenancy, Disaster Recovery (backing up and storing VMs)

List can go further like ... virtual environment for Dev Team for dev and test purposes → reduces time to market

# ISSUES & CHALLENGES

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**Performance Overhead:** Hypervisor overhead, resource contention

**Security Concerns**

**Management Complexity**

**Licensing & Costs**



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# NFV & VNF

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**Network Functions Virtualization** : NFV is a framework that enables network functions, traditionally implemented in hardware, to be run as software applications on general-purpose servers. This approach offers greater flexibility, scalability, and cost-effectiveness compared to traditional hardware-based network functions.

Focus: NFV focuses on the overall architecture and framework for deploying and managing virtualized network functions. It involves the separation of network functions from proprietary hardware and their deployment on a virtualized infrastructure.

**Virtualized Network Functions** : VNFs are individual network functions that have been virtualized. They are software applications that perform specific network functions, such as routing, switching, firewalling, or load balancing.

VNFs are an essential component of NFV. They are the building blocks that are deployed on virtualized infrastructure to create various network services.